INVERSE FUNCTIONS

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Name: Solutions

Laws of Exponents. Let $a, b \neq 1$ be positive numbers. If x and y are any real numbers, then

$b^{x+y} = b^x b^y$ $\quad$ $b^{x-y} = \frac{b^x}{b^y}$ $\quad$ $(b^x)^y = b^{xy}$ $\quad$ $(ab)^x = a^x b^x$

Simplify the following expressions.

1. $\dfrac{4^{-3}}{2^{-2}}$

$$\frac{4^{-3}}{2^{-2}} = \frac{2^2}{4^3} = \frac{4}{4^3}$$
$$= \boxed{\frac{1}{16}}$$

2. $8^{4/3}$

$$8^{4/3} = (2^3)^{4/3}$$
$$= 2^4$$
$$= \boxed{16}$$

3. $x(3x^2)^3$

$$x(3x^2)^3 = x(3^3)(x^2)^3$$
$$= 27x\,x^6$$
$$= \boxed{27x^7}$$

4. $b^8(2b^4)$

$$b^8(2b^4) = \boxed{2b^{12}}$$

Laws of Logarithms. Let $a, b \neq 1$ be positive numbers. If x and y are positive numbers, then

$$\log_b(xy) = \log_b(x) + \log_b(y) \qquad \log_b(x^r) = r\log_b(x)$$

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y) \qquad \log_b(x) = \frac{\log_a(x)}{\log_a(b)}$$

Find the given logarithm.

5. $\log_9(1)$

$\boxed{0}$

6. $\log_9(9^8)$

$\boxed{8}$

7. $\log_9(9)$

$\boxed{1}$

8. $\log_7(1)$

$\boxed{0}$

9. $\log_7(49)$

$\log_7(49) = \log_7(7^2)$

$= \boxed{2}$

10. $\log_7\left(\frac{1}{49}\right)$

$\log_7\left(\frac{1}{7^2}\right) = \log_7(7^{-2})$

$= \boxed{-2}$

11. $\log_3\left(\frac{1}{27}\right)$

$\log_3\left(\frac{1}{27}\right) = \log_3(1) - \log_3(27)$

$= 0 - \log_3(3^3)$

$= \boxed{-3}$

12. $\log_{10}(\sqrt{10})$

$\log_{10}(10^{1/2}) = \boxed{\frac{1}{2}}$

13. $\log_5(0.2)$

$\log_5(0.2) = \log_5\left(\frac{2}{10}\right)$

$= \log_5\left(\frac{1}{5}\right)$

$= \boxed{-1}$

Expand the given expression.

14. $\log_5\left(\frac{x}{2}\right)$

$\log_5\left(\frac{x}{2}\right) = \boxed{\log_5(x) - \log_5(2)}$

15. $\log_3\left(x\sqrt{y}\right)$

$\log_3(x\sqrt{y}) = \log_3(x) + \log_3(y^{1/2})$

$= \boxed{\log_3(x) + \frac{1}{2}\log_3(y)}$

16. $\log_3(5a)$

$\log_3(5a) = \boxed{\log_3(5) + \log_3(a).}$

17. $\log_5\left(\frac{2a}{b}\right)$

$\log_5\left(\frac{2a}{b}\right) = \log_5(2a) - \log_5(b)$

$= \boxed{\log_5(2) + \log_5(a) - \log_5(b)}$

18. $\log_{10}\left((w^2z)^{10}\right)$

$\log_{10}\left((w^2z)^{10}\right) = \log_{10}(w^{20}z^{10})$

$= \log_{10}(w^{20}) + \log_{10}(z^{10})$

$= \boxed{20\log_{10}(w) + 10\log_{10}(z)}$

19. $\log_7\left(\frac{\sqrt[3]{wz}}{x}\right)$

$\log_7\left(\frac{\sqrt[3]{wz}}{x}\right) = \log_7\left((wz)^{1/3}\right) - \log_7(x)$

$= \frac{1}{3}\log_7(wz) - \log_7(x)$

$= \boxed{\frac{1}{3}\log_7(w) + \frac{1}{3}\log_7(z) - \log_7(x)}$

Combine the given expression.

20. $4\log_2(x) - \frac{1}{3}\log_2(x^2+1)$

$4\log_2(x) - \frac{1}{3}\log_2(x^2+1)$

$= \log_2(x^4) - \log_2\left(\sqrt[3]{x^2+1}\right)$

$= \boxed{\log_2\left(\frac{x^4}{\sqrt[3]{x^2+1}}\right)}$

21. $\log_{10}(5) + 2\log_{10}(x) + 3\log_{10}(x^2+5)$

$\log_{10}(5) + 2\log_{10}(x) + 3\log_{10}(x^2+5)$

$= \log_{10}(5) + \log_{10}(x^2) + \log_{10}\left((x^2+5)^3\right)$

$= \boxed{\log_{10}\left(5x(x^2+5)^3\right)}$

22. $2\log_8(x+1) + 2\log_8(x-1)$

$2\log_8(x+1) + 2\log_8(x-1)$

$= 2\left(\log_8(x+1) + \log_8(x-1)\right)$

$= 2\log_8((x+1)(x-1))$

$= 2\log_8(x^2-1)$

$= \boxed{\log_8\left((x^2-1)^2\right)}$

23. $\log_5(x^2-1) - \log_5(x-1)$

$\log_5(x^2-1) - \log_5(x-1)$

$= \boxed{\log_5\left(\frac{x^2-1}{x-1}\right)}$